

Baseline-Guided Representation Learning for Noise-Robust EEG Signal Classification

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Abstract—Brain-computer interfaces (BCIs) suffer from limited accuracy due to noisy electroencephalography (EEG) signals. Existing denoising methods often remove artifacts such as eye movement or use techniques such as linear detrending, which inadvertently discard crucial task-relevant information. To address this issue, we present BGNet, a novel deep learning framework that leverages underutilized baseline EEG signals for dynamic noise mitigation and robust feature extraction to improve motor imagery (MI) EEG classification. Our approach employs data augmentation to strengthen model robustness, an autoencoder to extract features from baseline and MI signals, a feature alignment module to separate specific task and noise, and a classifier. We achieve state-of-the-art performance, an improvement of 5.9% and 3.7% on the BCIC IV 2a and 2b datasets, respectively. The qualitative analysis of our learned features proves superior representational power over baseline models, a critical aspect in dealing with noisy EEG signals. Our findings demonstrate the efficacy of readily available baseline signals in enhancing performance, opening possibilities for simplified BCI systems in brain-based communication applications.

Index Terms—brain-computer interface, electroencephalography, denoising, representation learning.

I. INTRODUCTION

Electroencephalography (EEG) classification holds immense potential for brain-computer interfaces (BCIs) [1], but its performance is limited by signal contamination from the environment and physiological artifacts such as eye and muscle movements [2]. These artifacts significantly impede accurate feature extraction from the weak target signal, thereby hindering downstream tasks [3]. Recognizing this limitation, previous studies have investigated denoising techniques using

electrooculography (EOG) and electromyography (EMG) signals to increase the signal-to-noise ratio [4]. However, their narrow focus on targeted artifacts, and reliance on linear detrending and filtering techniques [5], result in inadvertent removal of crucial task-relevant information.

In the field of motor imagery (MI) EEG classification, feature extraction from spatial, spectral, and temporal domains remains a challenge [6]. DCN [7] and EEGNet [8] are two prominent deep learning models that aim to extract MI features from noisy signals, without using artifact data. DCN proposes a deep convolutional neural network (CNN) to capture spectral power modulations while inherently considering spatial and temporal information. In contrast, EEGNet proposes a compact architecture that prioritizes learning frequency-specific spatial filters for each temporal filter. While both models establish significant performance, the limited availability of MI data poses challenges for training deep CNNs [9]. This raises concerns about the generalizability of extracted features and the risk for overfitting to dominate observed performances [10], [11]. Apart from using EOG and EMG signals, few studies leverage baseline EEG signals for denoising MI EEG signals despite their potential to reveal valuable noise information [12]. This can be attributed to the inherent complexity of baseline signal as it encompasses a wide range of brain activity patterns, making targeted denoising techniques challenging [13].

To address these challenges, we propose BGNet that utilizes baseline signals for denoising effect to improve subject-dependent MI classification. Our framework employs a feature alignment module for noise suppression by converging baseline features while separating MI features, and a two-stage training process to achieve robust model performance. Pre-training extracts diverse features using data augmentation, an autoencoder and the feature alignment module, and fine-tuning focuses on task-relevant discrimination using the autoencoder, feature alignment module, and a classifier.

In summary, this paper presents three key contributions:

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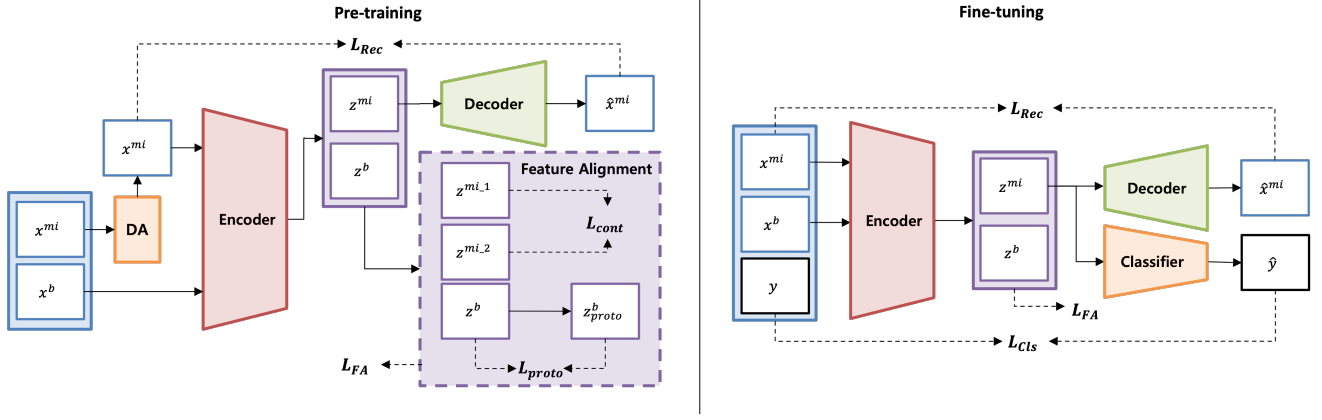


Fig. 1. Overall proposed framework of two-stage training process consists of data augmentation DA , an encoder to map signals x into latent space z , a decoder to reconstruct the signals $\hat{x}$ from z , a feature alignment module which effectively separates features in z , and a classifier to predict y . Pre-training is first employed to obtain a robust understanding of the signals, followed by fine-tuning for the specific classification task.

- Our feature alignment module and two-stage training process utilize baseline signals to denoise and decode MI signals. In pre-training, data augmentation enhances overall representation. In pre-training and fine-tuning, the autoencoder establishes a shared feature space, enabling the feature alignment module to effectively separate features for classification.
- Our model outperforms the baseline models on two benchmark datasets, demonstrating the effectiveness of our proposed framework.
- Unlike baseline models, our approach achieves spatial convergence of baseline features, and distinct and compact clusters for MI classes, highlighting superior feature representation of our proposed framework.

II. METHODOLOGY

In this section, we introduce BGNet that employs a feature alignment module to disentangle baseline and MI features, and a two-stage training process (pre-training and fine-tuning) to achieve robust model performance, as detailed in Fig. 1.

A. Notations

Let $\{(x_i^b, x_i^{mi}), y_i\}_{i=1}^N$ be N pairs of baseline and MI signals with its corresponding MI class label, where N denotes the total number of trials for a subject. When M replaces N , it denotes mini-batch size.

B. Pre-training

To mitigate the limited data issue for deep learning and enhance the diversity of MI data, inspired by Mixup [14], we create augmented signals in each mini-batch:

$$x_i^{mi'} = \alpha_i x_k^b + (1 - \alpha_i) x_i^{mi}, \quad i = 1, 2, \dots, M, \quad (1)$$

where α_i is a random value between 0 and 0.5 indicating the mixing weight, and k is a random number between 1 and M .

The augmented signals replace the original MI signals in the mini-batch, and serves as input to an autoencoder, designed for

efficient data representation [15], to extract task-relevant and noise features. This autoencoder employs a mapping function $E : x \rightarrow z$ to project signals into a lower-dimensional latent space, and a reconstruction function $D : z \rightarrow \hat{x}$, focusing on minimizing the reconstruction loss $\mathcal{L}_{Rec}$:

$$\mathcal{L}_{Rec} = \frac{1}{M} \sum_{i=1}^M (x_i^{mi'} - \hat{x}_i^{mi'})^2. \quad (2)$$

Simultaneously, our feature alignment module guides the autoencoder training by separating z^b and $z^{mi'}$. It aligns baseline features by minimizing a prototype loss $\mathcal{L}_{proto}$, which acts as a regularizer [16]:

$$\mathcal{L}_{proto} = \frac{1}{M} \sum_{i=1}^M \|z_{proto}^b - z_i^b\|_2^2, \quad (3)$$

where $z_{proto}^b = \frac{1}{N} \sum_{i=1}^N z_i^b$, and $\|\cdot\|_2$ denotes the Euclidean norm. It further discriminates features from different MI classes (mi'_1, mi'_2) by minimizing a contrastive loss $\mathcal{L}_{cont}$, which has been proven effective in feature learning [17]:

$$\mathcal{L}_{cont} = \frac{1}{P} \sum_{i=1}^P \max(0, \cos(z_i^{mi'_1}, z_i^{mi'_2})), \quad (4)$$

where P represents the number of pairs generated through the Cartesian product of z from different MI classes, and $\cos(\cdot)$ denotes the cosine similarity. The combined feature alignment module loss $\mathcal{L}_{FA}$ is as follows:

$$\mathcal{L}_{FA} = \mathcal{L}_{proto} + \mathcal{L}_{cont}. \quad (5)$$

The overall pre-training loss $\mathcal{L}_{Pre-train}$ fosters robust feature learning by minimizing:

$$\mathcal{L}_{Pre-train} = \mathcal{L}_{Rec} + \mathcal{L}_{FA}. \quad (6)$$

TABLE I
OVERALL CLASSIFICATION F1-SCORE ON THE BCIC IV 2A AND 2B DATASETS. BOLD INDICATES THE HIGHEST F1-SCORE.
WILCOXON SIGNED-RANK TEST. (*: $p < 0.05$, **: $p < 0.01$.)

Dataset	Model	Alignment ¹	S01	S02	S03	S04	Subject S05	S06	S07	S08	S09	Mean ↑ (Std ↓)
2a	DCN [7]	✗	55.47	33.47	62.90	38.08	26.36	33.70	46.95	51.07	54.49	*44.72 (11.03)
	EEGNet [8]	✗	64.12	37.78	72.96	38.02	27.40	32.10	56.29	71.92	69.98	52.29 (16.50)
	MIN2Net [18]	✓	38.78	33.09	45.59	36.22	60.40	33.50	59.75	31.05	32.40	*41.20 (10.34)
	BGNet ⁻	✗	50.82	34.23	57.08	34.35	25.49	31.86	43.38	54.34	53.29	*42.76 (10.37)
	BGNet ⁻ w/ FA	✓	29.49	30.81	51.88	33.56	24.65	31.80	34.84	41.44	48.64	**36.35 (08.15)
	BGNet (Ours)	✓	61.12	34.86	55.89	54.40	48.59	40.51	54.41	50.69	54.69	50.57 (07.33)
2b	DCN [7]	✗	75.35	51.68	54.80	95.93	87.01	81.87	78.12	77.62	79.90	*75.81 (12.71)
	EEGNet [8]	✗	77.58	52.57	55.11	95.80	87.81	83.26	80.94	81.73	82.50	77.48 (12.85)
	MIN2Net [18]	✓	71.39	51.37	86.61	47.97	85.87	43.35	88.95	88.09	56.23	68.87 (17.14)
	BGNet ⁻	✗	76.02	51.51	55.83	95.25	87.52	79.74	78.96	77.30	79.53	*75.74 (12.43)
	BGNet ⁻ w/ FA	✓	76.51	51.15	52.35	95.79	87.15	80.28	77.26	78.87	79.95	**75.48 (13.17)
	BGNet (Ours)	✓	76.15	59.26	55.67	98.51	89.31	78.31	89.69	81.36	86.93	79.47 (12.69)

¹ The process of aligning features in the latent space.

C. Fine-tuning

After pre-training, we transition to focus on task-relevant discrimination. During fine-tuning, data augmentation is omitted. The pre-trained autoencoder weights initialize a fine-tune autoencoder equipped with the feature alignment module and a classifier. The classifier's prediction task maps latent features to class labels using the function $C : z \rightarrow \hat{y}$, and is achieved by minimizing a cross-entropy loss $\mathcal{L}_{Cls}$:

$$\mathcal{L}_{Cls} = \frac{1}{M} \sum_{i=1}^M \sum_{j=1}^L -y_{ij} \log \frac{\exp(\hat{y}_{ij})}{\sum_{k=1}^L \exp(\hat{y}_{ik})}, \quad (7)$$

where L is the number of MI classes.

The overall fine-tuning loss $\mathcal{L}_{Fine-tune}$ jointly optimizes both autoencoder and classifier, which improves training stability and model's generalization performance by minimizing:

$$\mathcal{L}_{Fine-tune} = \mathcal{L}_{Rec} + \mathcal{L}_{FA} + \mathcal{L}_{Cls}. \quad (8)$$

III. EXPERIMENTS

A. Datasets

The BCIC IV 2a [19] dataset features 22-channel EEG across 4 MI tasks (left hand, right hand, feet, and tongue) from 9 subjects with 144 trials/task/subject. The BCIC IV 2b [19] dataset offers 3-channel EEG, focusing on left hand and right hand tasks from 9 other subjects with 360 trials/task/subject. Both datasets, at a 250 Hz sampling rate, use cue-based trials with fixation, cue and execution, and rest periods. We extract baseline signals from fixation, and 3-second MI signals from cue and execution periods for model training.

B. Models

DCN [7], EEGNet [8] and MIN2Net [18] are identified for model comparison. MIN2Net employs triplet loss to align only MI features in the latent space. We adopt a DCN-based autoencoder (BGNet⁻), and conduct an ablation study to justify the benefits of our feature alignment module (BGNet⁻ w/ FA), and our proposed model (BGNet).

C. Setups

We apply a sixth-order Butterworth filter between 0.1 and 40 Hz to the signals. Models are trained across 5-fold cross-validation for a maximum of 500 epochs, with Adam optimizer, learning rate of 10^{-4} and batch size of 16. A single linear layer is employed for classification. Training stops after 10 epochs without validation improvement to prevent overfitting with a minimum of 200 epochs.

IV. RESULTS

A. Quantitative Analysis

Table I delineates the subject-specific, and overall mean subject classification F1-scores, along with the standard deviations of different models on the two benchmark datasets. Comparing to DCN, our model observes a 5.9% and 3.7% performance increase on the BCIC IV 2a and 2b datasets, respectively. We achieve the highest F1-score of 79.47% on the 2b dataset, which is approximately 2% higher than the second best EEGNet, highlighting the capability of our framework in learning accurate representations. While our model of 50.57% yields a less than 2% difference compared with the best EEGNet of 52.29%, it achieves a much lower standard deviation. This suggests a less robust training for the deep learning models, possibly due to the combination of a smaller data size and a larger number of classes.

Furthermore, BGNet⁻ performs on par with DCN. Comparing to MIN2Net, findings suggest that employing our feature alignment module in isolation (BGNet⁻ w/ FA) is insufficient to improve model performance. However, the combination of both our feature alignment module and two-stage training process (BGNet) proves successful, achieving significantly higher performance. Notably, BGNet demonstrates a 9.4% and 10.6% increase in performance on the 2a and 2b datasets, respectively, offering compelling evidence for the effectiveness of our proposed method.

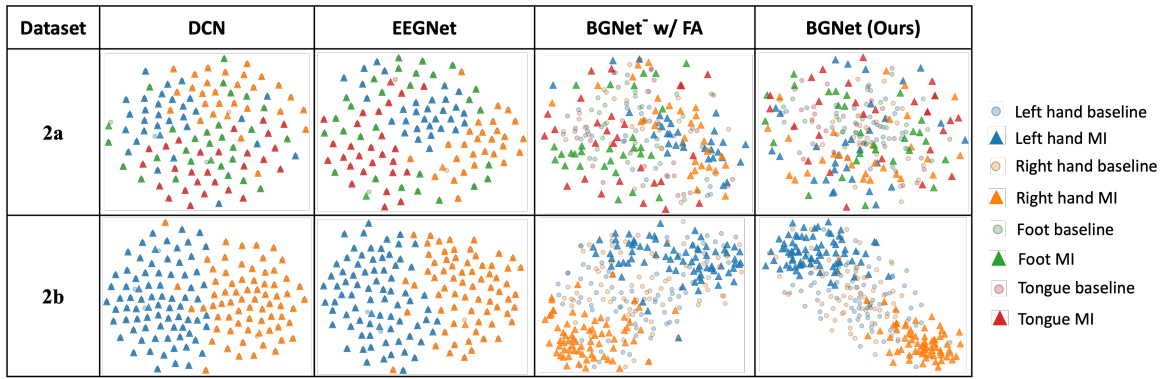


Fig. 2. t-SNE visualization of the BCIC IV 2a subject 3 and 2b subject 4 test datasets' baseline and MI features.

B. Qualitative Analysis

To analyze the representational power of the models, we select the best performing subject 3 and 4 of the BCIC IV 2a and 2b dataset, respectively, and visualize the extracted features using t-SNE. With reference to Fig. 2, DCN and EEGNet achieve MI class separation within the feature space. This differentiation, however, extends to baseline features. As baseline signals lack MI task-relevant information, this finding suggests reliance on noise features for class discrimination, potentially confounding genuine MI features. Conversely, our models show otherwise, where baseline and MI features do not overlap. In the attempt to learn the class-discriminative latent space, our models are able to resist overfitting to noise. Notably, the result of BGNet on the 2b dataset exhibits high class-separability for MI, and not for baseline signals. This proves that our models can effectively extract genuine task-relevant features from the noisy MI signals, which are not present in the baseline signals. In addition, the feature visualization results demonstrate the potential flaws of existing classification methods, and the usefulness of our methodology in EEG representation learning beyond mere classification.

V. CONCLUSION

We propose BGNet, a subject-dependent representation learning framework that utilizes baseline EEG signals for dynamic noise mitigation and robust feature extraction, for MI classification. Our proposed framework employs data augmentation to strengthen model robustness, an autoencoder to extract features from baseline and MI signals, a feature alignment module to separate baseline noise and specific task, and a classifier. We achieve substantial performance improvements compared to state-of-the-art models on two benchmark datasets, and demonstrate our model's effectiveness in preserving the integrity of MI EEG signals. These findings highlight the inherent richness of baseline signals, previously underutilized in BCI research, as a source of insight into intrinsic noise patterns. Our work eliminates the need for additional artifact data to denoise signals, paving the way for a new era of accessible and efficient EEG applications in neurofeedback and brain-based communication.

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